

M.A.P.S
Multiphoton Absorption Prediction Software
Manual
Version 2026-1

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Contents

1	Overview to M.A.P.S	2
2	Downloading and Using M.A.P.S	3
2.1	Applet Setup	3
2.1.1	Downloading from GitHub	3
2.2	Select Log Directory Popup	3
3	Navigating the Home Page	4
4	Modeling Excitation Processes	9
4.1	Output 1	10
4.2	Output 2	10
4.3	Output 3	11
5	Modeling Dissociation Processes	12
5.1	Output 1	13
5.2	Output 2	13
6	Modeling Ionization Processes	14
6.1	Ionization Figure	15
7	Managing the Output Plots	15
7.1	The Header Bar	15
7.2	File Menu	16
7.3	Hover Tools on the Plot	16
7.4	Summary	17
7.5	Wrapping Up	17

1 Overview to M.A.P.S

M.A.P.S (Multiphoton Absorption Prediction Software) is an interactive GUI tool designed to simulate the effects of multiphoton absorption (MPA) and multiphoton ionization (MPI) on biological samples. The software provides a user-friendly way to explore how nonlinear light-matter interactions impact chemical molecules during high-fluence pump-probe experiments.

This program supports the findings presented in the paper *The role of multiphoton absorption and ionization in structural dynamic studies* by Alexander Wainwright *et al.* (add in other contributors here) at the Miller Group and Atomically Resolved Dynamics in the University of Toronto. In this work, a predictive model was developed to identify the excitation fluence regimes where MPA and MPI become significant in time-resolved serial femtosecond crystallography (TR-SFX) experiments. The model was benchmarked against published datasets of carboxymyoglobin (MbCO) dissociation and the bacteriorhodopsin (Br) photocycle, providing quantitative thresholds for the onset of nonlinear pathways. These findings offer practical guidance for optimizing excitation parameters so that observed structural changes reflect biologically meaningful dynamics.

Explicit Usage Note: *M.A.P.S is intended as a predictive tool only when experimental constants are already known. If such constants are not available, M.A.P.S should instead be used as a consistency or sanity check, rather than a standalone experimental predictor. In practice, at least one experiment must be conducted to obtain the necessary constants before M.A.P.S can provide predictive outputs.*

Development Background and Credits

M.A.P.S was developed between June–August 2025 by an undergraduate student researcher in the Miller Group and the Atomically Resolved Dynamics group at the University of Toronto.

- The core models against which this software is benchmarked were written by Alexander Wainwright, following prior modeling of multiphoton effects in bacteriorhodopsin and MbCO.
- The GUI development, user input customization, and generalization of the models (originally limited to MbCO and bacteriorhodopsin) were carried out by Syeda Mahdia.
- Additional features such as vectorized calculations and split rate-equation modeling were also incorporated by Syeda Mahdia to improve computational efficiency.
- Claude AI was used to support GUI figure and text layout, as well as in places where it is cited directly as the author in the code comments.

For questions, queries and qualms about the software, please contact:

- snmahdia@gmail.com or syeda.mahdia@mail.utoronto.ca

Please refer to the related paper for further details: *Alexander A.C. Wainwright, Syeda N. Mahdia, Khaled Madhoun, Jessica E. Besaw, R.J. Dwayne Miller, Modeling Multiphoton Absorption and Ionization in Ultrafast Pump-Probe Experiments, JCP (Under Review), 2026.*

2 Downloading and Using M.A.P.S

2.1 Applet Setup

2.1.1 Downloading from GitHub

To install M.A.P.S:

1. Go to the GitHub repository: <https://github.com/17aacw/MPA-Rate-Equations/tree/main/GUI>.
2. Click the green “Code” button and select Download ZIP.
3. Extract the downloaded ZIP file into a directory of your choice.
4. Open the extracted folder and follow the instructions in the `README.md`

Note: If you are using a school computer and encounter permission issues, please contact your system administrator or ask a colleague with higher-level permissions to perform the installation.

Tip: Save the app to your desktop or pin it to your taskbar for easier access. If you are on a Windows you are using a school computer and encounter permission issues, please contact your system administrator or ask a colleague with higher-level permissions to perform the installation.

2.2 Select Log Directory Popup

Log Directory Selection This section walks through the Select Log Directory Popup which precedes entering the simulation software’s home page.

After starting the applet, and once the splash screen has finished, a popup window titled M.A.P.S - Select Log Directory will appear.

When running the app for the first time, allow some extra time for the program to install and configure its dependencies. This setup should happen automatically, *without the need for installing MATLAB*.

M.A.P.S generates a log file in .txt format that records simulation inputs, outputs, calculation updates, computation time, and final results. This file is saved together with the graphs produced by the program so that all parameters used to create the graphs are documented.

The popup window allows you to choose where the log file is saved. By default, it is stored in the current directory where the app was installed, but you may also select a different folder on your computer.

After selecting a directory, the popup updates as follows:

- The label Current Directory changes to Selected Directory.
- A green status message is displayed.
- A “Continue” button appears.

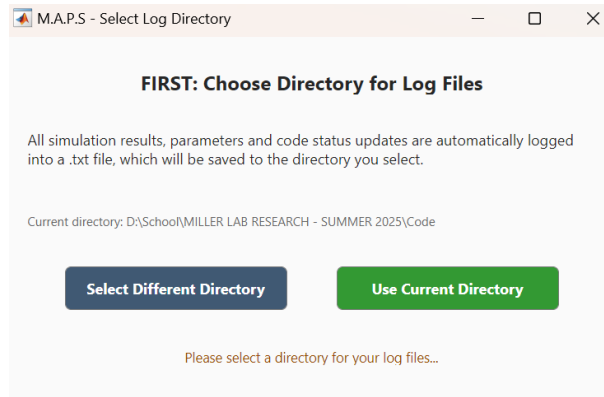


Figure 1: *Initial Popup Window*

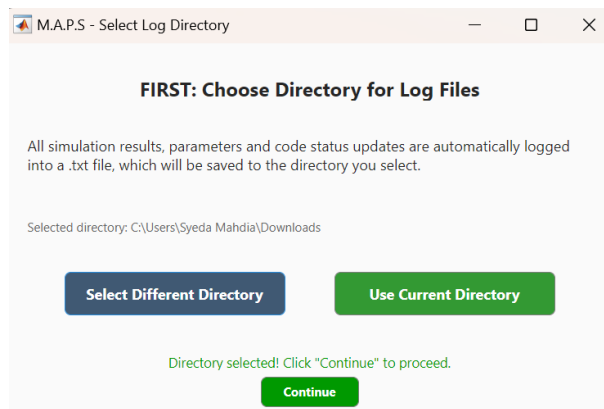


Figure 2: *Updated Popup Window*

Click the “Continue” button to proceed to the M.A.P.S Home Page.

3 Navigating the Home Page

After the log directory is selected, users are directed to the main simulation page, which by default opens to the excitation model.

Multi-Photon Effects

MPA Excitation, Dissociation & Ionization

PROCESS: Excitation **Name of Sample:** My Sample

Input Variables

Protein Concentration (M)	0.03		
Absorption Coefficient α (cm^{-1})	1e+04		
σ_{01} (cm^2)	1e-20		
Photons per Chromophore (Calc.)	0		
Wavelength (nm)	530	Pulse Duration (s)	2e-13
Laser Peak Power (W/cm^2)	1e+13	σ_{0n} ($\text{m}^4/\text{s}/\text{photon}$)	2.9e-56
σ_{1n} (cm^2)	1e-20	Dephasing Time S1 (s)	2.8e-14
Decay Lifetime S1 (s)	4.5e-13	Decay Lifetime S4 (s)	2e-13
Max depth (m)	3e-06		

ODE Solver

☒ ode45 (non-stiff)
 ☐ ode23 (non-stiff)
 ☐ ode15s (stiff)
 ☐ ode23s (stiff)
 ☐ Euler (explicit)

Run Simulation Close Plots

Figure 3: Home Page Initial View

On this page, users will see the title of the software and a dropdown menu labeled Process. This menu allows selection of the type of process to be modeled.

Multi-Photon Effects

MPA Excitation, Dissociation & Ionization

PROCESS: Excitation **Name of Sample:** My Sample

Input Variables

Protein Concentration (M)	0.03		
Absorption Coefficient α (cm^{-1})	1e+04		
σ_{01} (cm^2)	1e-20		
Photons per Chromophore (Calc.)	0		
Wavelength (nm)	530	Pulse Duration (s)	2e-13
Laser Peak Power (W/cm^2)	1e+13	σ_{0n} ($\text{m}^4/\text{s}/\text{photon}$)	2.9e-56
σ_{1n} (cm^2)	1e-20	Dephasing Time S1 (s)	2.8e-14
Decay Lifetime S1 (s)	4.5e-13	Decay Lifetime S4 (s)	2e-13
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 ☐ Euler (explicit)

Run Simulation Close Plots

Figure 4: Process Menu

The available processes in this version are excitation, dissociation, and ionization.

Multi-Photon Effects

MPA Excitation, Dissociation & Ionization

PROCESS: **Excitation** Name of Sample: My Sample

Input Variables

Protein Concentration (M)	0.03		
Absorption Coefficient α (cm^{-1})	1e+04		
σ_{01} (cm^2)	1e-20		
Photons per Chromophore (Calc.)	0		
Wavelength (nm)	530	Pulse Duration (s)	2e-13
Laser Peak Power (W/cm^2)	1e+13	σ_{0n} ($\text{m}^4\text{-s}/\text{photon}$)	2.9e-56
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 ☐ ode23s (stiff)
 ☐ Euler (explicit)

Run Simulation Close Plots

Figure 5: *Process Menu Dropdown*

Details on how these processes are modeled can be found in the referenced paper, where the underlying rate equation sets are presented.

Next to the dropdown menu is a field labeled Name of Sample. This field customizes the titles of the figures generated by M.A.P.S and also determines the default name of the simulation log file.

Multi-Photon Effects

MPA Excitation, Dissociation & Ionization

PROCESS: Excitation Name of Sample: **My Sample**

Input Variables

Protein Concentration (M)	0.03		
Absorption Coefficient α (cm^{-1})	1e+04		
σ_{01} (cm^2)	1e-20		
Photons per Chromophore (Calc.)	0		
Wavelength (nm)	530	Pulse Duration (s)	2e-13
Laser Peak Power (W/cm^2)	1e+13	σ_{0n} ($\text{m}^4\text{-s}/\text{photon}$)	2.9e-56
σ_{1n} (cm^2)	1e-20	Dephasing Time S1 (s)	2.8e-14
Decay Lifetime S1 (s)	4.5e-13	Decay Lifetime S4 (s)	2e-13
Max depth (m)	3e-06		

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 ☐ ode23s (stiff)
 ☐ Euler (explicit)

Run Simulation Close Plots

Figure 6: *Name of Sample*

A button labeled Change Log Directory is also available. This allows users to change the folder where simulation log files are stored.

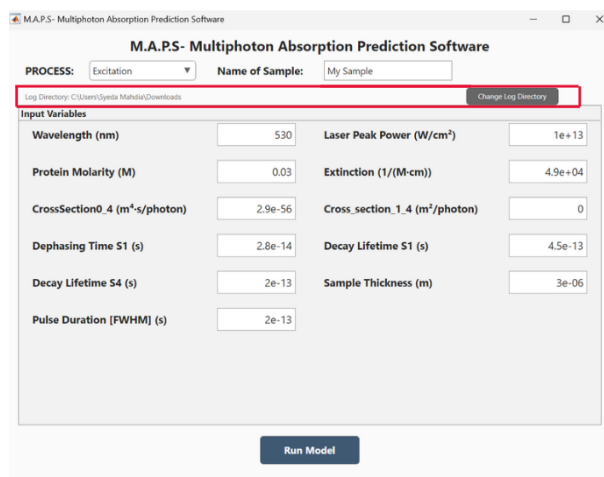


Figure 7: *Change Log Directory Button & Current Log Directory*

To select the ODE solver please click the appropriate radial button in the ODE Solver box.

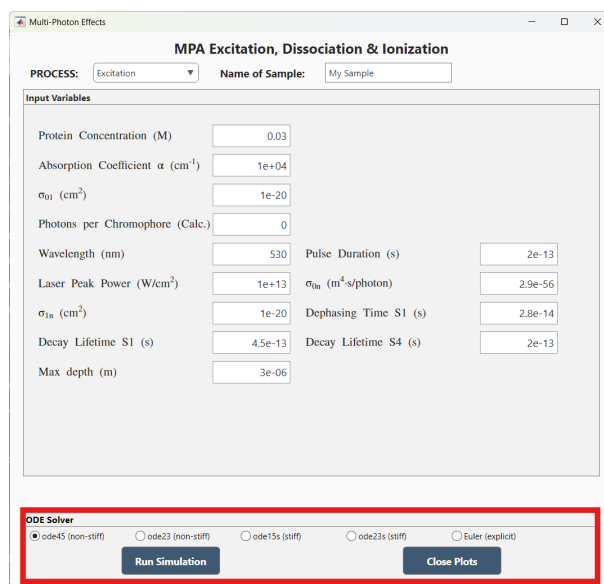


Figure 8: *To change your solver, select the radial button in this box. The default is ODE45, which was used in the associated manuscript*

To close the plots from the previous model run please click Close Plots button.

Multi-Photon Effects

MPA Excitation, Dissociation & Ionization

PROCESS: Excitation Name of Sample: My Sample

Input Variables

Protein Concentration (M)	0.03		
Absorption Coefficient α (cm^{-1})	1e+04		
σ_{01} (cm^2)	1e-20		
Photons per Chromophore (Calc.)	0		
Wavelength (nm)	530	Pulse Duration (s)	2e-13
Laser Peak Power (W/cm^2)	1e+13	σ_{0n} ($\text{m}^2\cdot\text{s}/\text{photon}$)	2.9e-56
σ_{1a} (cm^2)	1e-20	Dephasing Time S1 (s)	2.8e-14
Decay Lifetime S1 (s)	4.5e-13	Decay Lifetime S4 (s)	2e-13
Max depth (m)	3e-06		

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 ☐ ode15s (stiff)
 ☐ ode23s (stiff)
 ☐ Euler (explicit)

Run Simulation
Close Plots

Figure 9: To close the plots from your previous simulation press the close plots button.

To Run your simulation or model please click Run Simulation button.

Multi-Photon Effects

MPA Excitation, Dissociation & Ionization

PROCESS: Excitation Name of Sample: My Sample

Input Variables

Protein Concentration (M)	0.03		
Absorption Coefficient α (cm^{-1})	1e+04		
σ_{01} (cm^2)	1e-20		
Photons per Chromophore (Calc.)	0		
Wavelength (nm)	530	Pulse Duration (s)	2e-13
Laser Peak Power (W/cm^2)	1e+13	σ_{0n} ($\text{m}^2\cdot\text{s}/\text{photon}$)	2.9e-56
σ_{1a} (cm^2)	1e-20	Dephasing Time S1 (s)	2.8e-14
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Max depth (m)	3e-06		

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 ☐ ode15s (stiff)
 ☐ ode23s (stiff)
 ☐ Euler (explicit)

Run Simulation
Close Plots

Figure 10: To run your simulation press the run simulation button.

At the bottom of the page, there is a darker gray Input Variables section. This area contains the parameters used for the simulations. By default, it is pre-populated with the values used in the benchmark simulations for bacteriorhodopsin and MbCO.

Figure 11: *Advanced options for dissociation simulations.*

Specifically, the pre-populated values depend on the selected process. For excitation and ionization, the default parameters are based on bacteriorhodopsin, as described in the referenced paper. For dissociation, the default parameters are derived from MbCO. The details of how these values were obtained are thoroughly explained in the accompanying paper linked to this simulation.

4 Modeling Excitation Processes

The excitation simulation screen provides the following input parameters:

- **Wavelength:** The wavelength of the XFEL or laser source used in the simulation (in nm).
- **Protein Molarity:** Concentration of the protein sample (in mol).
- **Dephasing Time (S_1):** Time constant for electronic dephasing (in seconds).
- **Decay Lifetime (S_4):** Time constant for population decay (in seconds).
- **Pulse Duration:** Duration of the laser pulse, defined as the full width at half maximum (FWHM) in seconds.
- **Laser Peak Power:** Maximum power of the laser pulse.

These parameters can be modified as needed for the sample under study.

Expected Outputs

Running the excitation model produces the following.

- A log file recording input values, diagnostic calculations, and results.

- Numerical outputs describing excitation dynamics.
- Graphs showing the excitation behavior based on the chosen parameters.

There will be 3 graph outputs produced by the excitation model:

4.1 Output 1

This plot shows the population dynamics of the main electronic states (S0–S4) of bacteriorhodopsin under multiphoton excitation.

The vertical axis represents the fractional population of each state, while the horizontal axis is time.

Initially, most molecules reside in the ground state (S0). Upon multiphoton excitation, population is transferred to higher excited states (S1–S4).

The curves illustrate how the populations evolve with time, indicating excitation, relaxation, and possible competing pathways.

A key point of analysis is how quickly S1 and S2 states rise and decay, since they govern the subsequent photochemical events.

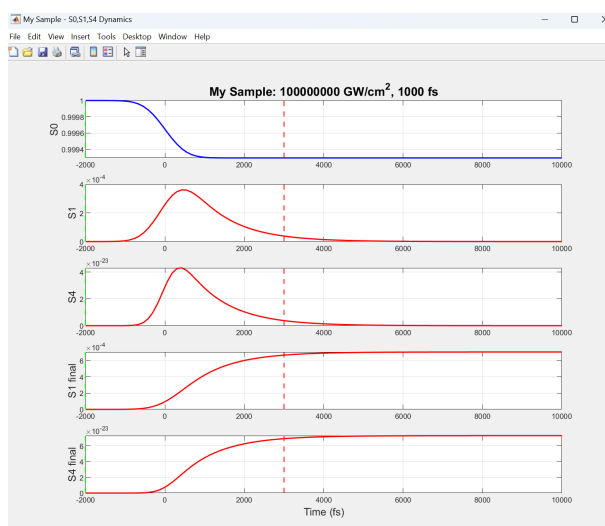


Figure 12: *Excitation Graph 1- S0, S1, S4 Dynamics*

4.2 Output 2

This plot provides a more detailed look at energy-level interactions.

It tracks the relative occupation of key vibrational or intermediate levels within the electronic states.

This output is especially useful for diagnosing whether the multiphoton excitation leads to accumulation in higher levels or rapid relaxation.

Researchers can use this graph to verify that the chosen laser parameters produce the desired excitation pathway without trapping population in unwanted states.

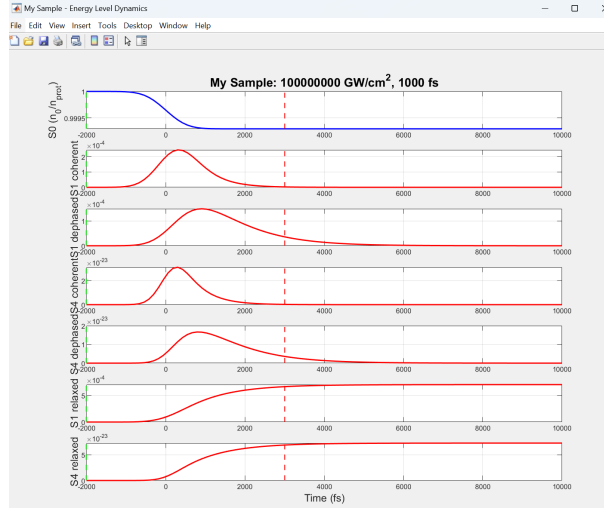


Figure 13: *Excitation Graph 2- Energy Level Dynamics*

4.3 Output 3

This plot compares simulation results for excitation yield and the relative contribution of two-photon absorption (2PA) versus intermediate photon absorption (IPA).

The top graph shows overall excitation yield as a function of time or input intensity, helping to quantify efficiency.

The lower colormap illustrates the ratio of 2PA to IPA under different excitation conditions, providing a visual guide to which mechanism dominates.

Interpreting this plot is crucial for optimizing laser parameters: a higher 2PA contribution suggests genuine multiphoton effects, while a larger IPA ratio may indicate stepwise absorption dominating.

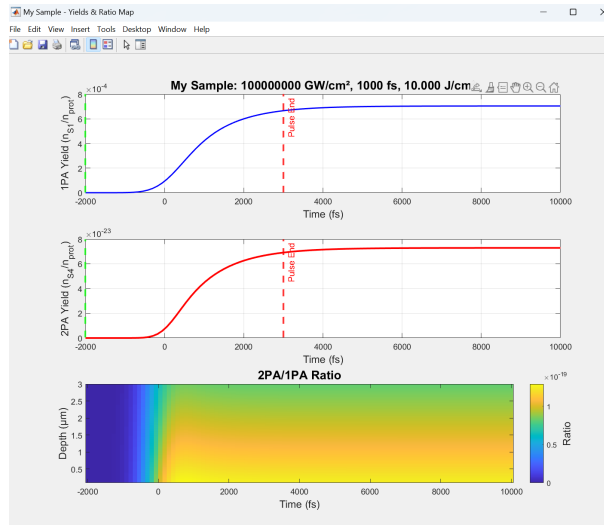


Figure 14: *Excitation Graph 3- Yield and 1PA vs 2PA ratio*

5 Modeling Dissociation Processes

The dissociation simulation screen provides the following input parameters:

- **Cross Section σ_{01} :** One-photon absorption cross section between the ground state and the first excited state (in m^2).
- **Cross Section σ_{12} :** One-photon absorption cross section between the first and second excited states (in m^2).
- **Cross Section σ_{02} :** Two-photon absorption cross section for direct ground-to-second excited state transitions (in $\text{m}^4 \cdot \text{s} / \text{photon}$).
- **Peak Intensity:** Peak laser intensity applied to the sample (in W/cm^2).
- **Pulse Duration τ_p :** Duration of the laser pulse, defined as the full width at half maximum (FWHM) (in fs).
- **Relaxation Time τ_1 :** Relaxation time constant from the first excited state (in fs).
- **Relaxation Time τ_2 :** Relaxation time constant from the second excited state (in fs).
- **Wavelength:** Central wavelength of the laser source (in nm).
- **Sample Depth:** Physical thickness of the sample being irradiated (in m).
- **Absorption Coefficient α :** Linear absorption coefficient of the material (in m^{-1}).

Advanced Input:

- **Spatial Resolution N_z :** Number of spatial points used in the simulation along the sample depth, which controls how finely the pulse interaction with the medium is resolved.

These parameters can be adjusted to match the sample and laser setup for dissociation modeling.

Expected Outputs

Running the dissociation model produces the following.

- A log file recording input values, calculations, and results.
- Numerical outputs describing dissociation dynamics.
- Graphs showing dissociation behavior based on the chosen parameters.

There are 2 graph outputs produced by the dissociation model:

5.1 Output 1

This plot shows how the populations of bound and dissociated states change over time.

The vertical axis is the fractional population and the horizontal axis is time.

At the start, most molecules remain in the bound state. As the simulation runs, population shifts into dissociation channels.

The curves indicate how dissociation grows and how competing pathways reduce or slow it down.

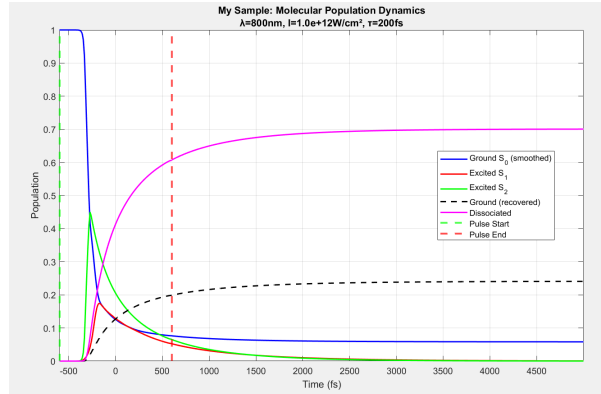


Figure 15: *Dissociation Population Dynamics Modeling*

5.2 Output 2

This plot compares dissociation from one-photon and two-photon processes.

The vertical axis is the fractional population from each pathway and the horizontal axis is time.

Solid lines show two-photon contributions, while dashed lines show single-photon contributions.

This lets the user see which pathway dominates under the chosen parameters and compare the efficiency of one- vs two-photon dissociation.

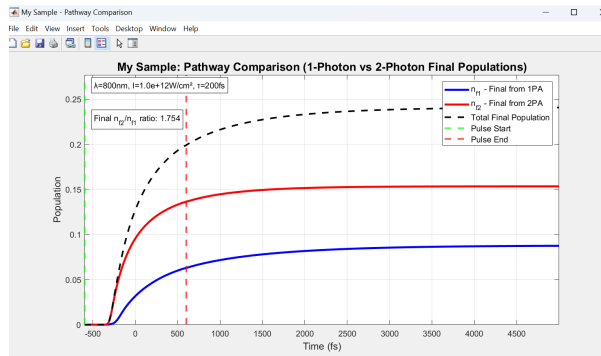


Figure 16: *Dissociation Graph 2 - Pathway Comparison between 1 and 2 Photon Populations*

6 Modeling Ionization Processes

The ionization simulation screen provides the following input parameters:

- **Wavelength:** Central wavelength of the laser source (in m).
- **Pulse Duration τ_p :** Duration of the laser pulse, defined as the full width at half maximum (FWHM) (in s).
- **Spot Size:** Effective radius of the laser beam waist (in μm).
- **Peak Power FWHM:** Peak laser power per unit area at FWHM (in W/cm^2).
- **Bandgap Energy E_g :** Minimum photon energy required to excite electrons from the valence band to the conduction band (in eV).
- **Intermediate Level Energy:** Energy of defect or trap states in the bandgap that can play a role in multiphoton absorption (in eV).
- **Electron Collision Time τ_c :** Average time between electron scattering events in the medium (in s).
- **Bound Electron Density N_b :** Density of bound electrons available for ionization (in m^{-3}).
- **Initial Neutral Density N_0 :** Initial density of neutral atoms or molecules in the sample (in cm^{-3}).

Advanced Input:

- **Log Directory:** Location where the program stores all input parameters, results, and log files.

These parameters can be adjusted to match the specific material and laser setup for ionization modeling.

Expected Output

The ionization model produces one main result: how the electron density evolves over time as the laser pulse passes through the sample.

- A log file saves the inputs, calculations, and results.
- Numerical outputs include the total ionization and the relative contributions from avalanche ionization (AI) and strong-field ionization (SFI).
- A single plot is generated showing electron density versus time.

6.1 Ionization Figure

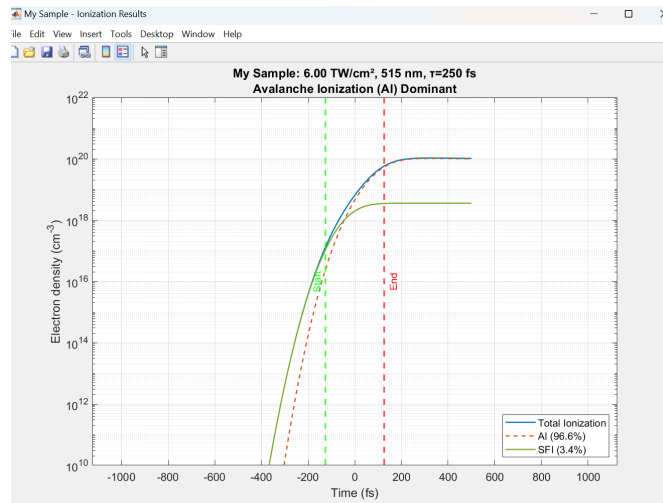


Figure 17: Ionization Output Plot

Figure 17 tracks electron density (cm^{-3}) over time (fs). In this specific case, the density is negligible, but once the pulse ramps up, ionization takes off quickly. In this example, avalanche ionization dominates, making up about 96.6% of the total electron density, while strong-field ionization contributes only 3.4%.

The blue curve shows the total ionization, the dashed red line is avalanche ionization, and the green curve is strong-field ionization. The plot title *updates automatically* to highlight which mechanism is dominant, making it easy to see both the main pathway and how fast ionization develops during the pulse. There are green and red pulse markers that mark the start and end of the plot.

7 Managing the Output Plots

Each time the software generates a new plot, it opens in its own popup window. These windows are designed to feel familiar if you've worked with MATLAB figures before: there's a header bar at the top with menus, and when you hover over the figure itself, a small set of quick-access icons appears. Below is a walkthrough of everything you'll see and what each option does.

7.1 The Header Bar

At the very top of the plot window is the header bar (Figure 18). This bar contains menus like **File**, **Edit**, and others. These menus are where you'll find most of the options for saving, exporting, and customizing your plots.

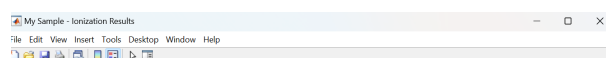


Figure 18: The header bar at the top of an output window.

7.2 File Menu

Clicking on **File** (Figure 19) drops down a menu with several choices. Here's a rundown of what each option does:

- **New Figure:** Open a blank figure window.
- **Open:** Load a previously saved figure file back into the viewer.
- **Save:** Save the current figure using its existing name and format.
- **Save As:** Save the figure under a new name or in a different format (PNG, JPEG, PDF, etc.).
- **Export Setup:** Adjust how the figure will look before exporting — for example, set resolution, fonts, or figure size.
- **Page Setup:** Configure layout settings if you plan to print (margins, orientation, etc.).
- **Print Preview:** See exactly how the figure will look when printed.
- **Print:** Send the figure directly to your printer.
- **Close:** Close this figure window without quitting the whole program.

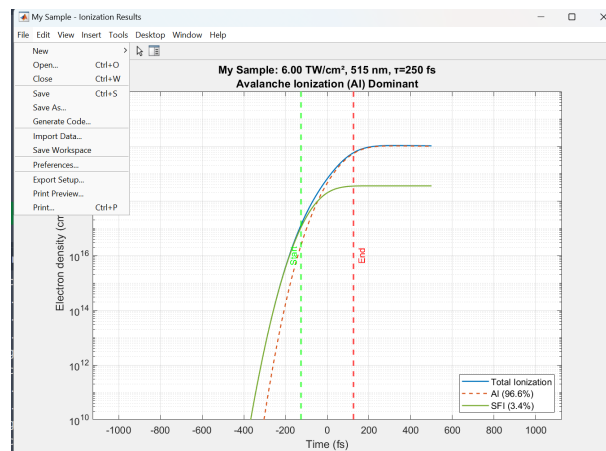


Figure 19: The File menu options.

7.3 Hover Tools on the Plot

If you move your mouse over the grid of the plot itself, a floating toolbar will appear (Figure ??). These tools let you interact with the plot directly, without going through menus:

- **Zoom In:** Focus on a specific region of the plot.
- **Zoom Out:** Step back to see more of the data.
- **Pan:** Click and drag to move the view around without changing the data.
- **Save Image:** Quickly save the current view of the plot in different file formats.

- **Edit Format:** Change labels, axis settings, and visual styles directly.
- **Add Notes / Comments:** Place annotations on the figure for future reference.
- **Home View:** Reset everything back to the original default view.

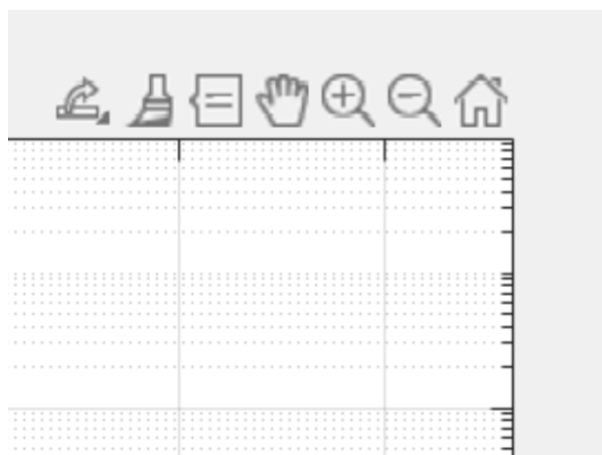


Figure 20: Pop-up Toolbar from Hovering over Figure

7.4 Summary

Between the drop-down menus in the header bar and the hover tools on the plot, you have full control over how to save, explore, and customize your figures. Everything you see in the interface is designed to make working with plots quick and flexible; whether you're zooming in for detail, printing out a hard copy, or dropping images into a report.

7.5 Wrapping Up

This is the last section of the manual, so you should be ready to dive in and use the software on your own. Don't be afraid to experiment with the tools as you go - the whole point is to make interacting with your results simple and flexible. Thanks for taking the time to read through, feel free to contact the dev for any questions, concerns, inquiries and feedback, and happy plotting!